

NETWORK CITY

COMPUTER HARDWARE

Meet the parts that power your PC



MEET THE PARTS. CHECK THE FIT. SOLVE THE SCENARIO.

Detailed comic illustrations + short definitions + clear exam clues

Explore the parts inside a PC, then tackle displays, cables, and printers. Learn definitions, acronym meanings, compatibility rules, and safety steps for Core 1 hardware objectives 3.1–3.8.

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Independent study guide. Aligned to exam 220-1201; not an official CompTIA publication. Illustrations are memory aids. Technical notes and labeled diagrams explain the real rules.

Meet the hardware team

VISUAL: Draw a neon computer city: gold CPU captain, green RAM workers, purple storage vault, and orange power station.



Different parts. One working computer!

BOOT!

- **Hardware.** The physical parts of a computer. You can touch a keyboard, circuit board, drive, or screen. Software is the set of instructions those parts run.

- **Motherboard.** The main circuit board. It connects the processor, memory, storage, power, and add-on devices. Think of the city roads and building sites.

- **CPU - Central Processing Unit.** The instruction worker. It carries out calculations and directs many computer tasks. Our gold captain needs the rest of the team to get work done.

- **RAM - Random-Access Memory.** The temporary worktable for active programs and data. Normal system RAM loses its contents when power is removed.

- **Storage.** The file cabinet for saved data. An SSD or HDD keeps files after shutdown. Save a document to storage; do not rely on its temporary copy in RAM.

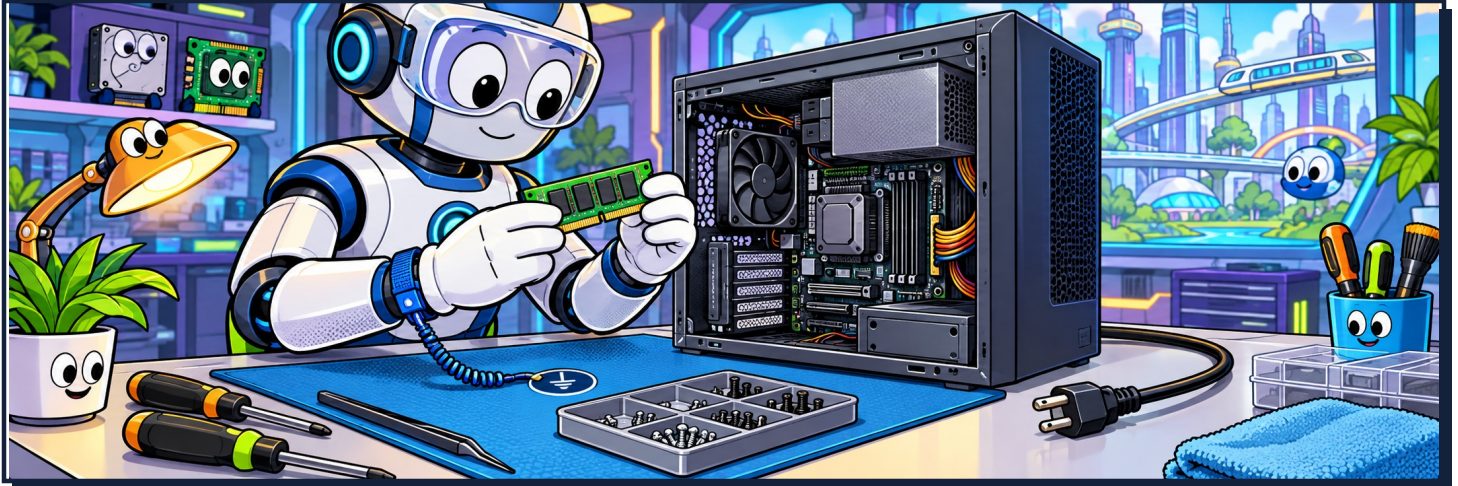
- **Input / output.** Input brings information in, such as a keyboard press. Output sends information out, such as a screen image or printed page. Some devices do both.

- **Driver / OS.** A driver helps an operating system, or OS, work with a device. The OS manages resources and provides services for apps.

SCOPE: Core 1, 220-1201, hardware objectives 3.1-3.8. Safety and fault clues support the lessons. This is a broad study guide, not a promise that every exam detail fits in one book. [1]

Safe hands before fast hands

VISUAL: Draw a careful robot at a grounded antistatic bench. The desktop is unplugged, and the robot holds a board by its edges.



Power down. Protect the parts. Read the manual!

CHECK!

- **ESD - Electrostatic Discharge.** A sudden flow of static electricity. A spark too small to feel can damage a chip. Use the approved grounding method for the service task. [2]

- **Before opening.** Shut down, disconnect external power, and isolate the battery when the service manual requires it. Follow the model's steps for draining residual system power.

- **Wrist strap / mat.** An antistatic strap and mat help keep you and the parts at the same electrical potential. Use approved grounding equipment; never improvise a connection to a live electrical contact.

- **Hold the edges.** Avoid touching chips, gold contacts, and socket pins. Keep parts in antistatic packaging until needed. Keep drinks away and track every screw.

- **Stored energy.** A power supply or old CRT, a cathode-ray tube display, can hold dangerous voltage. Do not open these assemblies during routine computer service. [3]

- **Hot and damaged parts.** Let heatsinks and printer fusers cool. Stop if you find burning smells, leaking parts, or a swollen battery. Do not puncture or crush a battery.

THE SAFETY RULE BEATS THE SPEED RULE. Follow the device manual and workplace procedure. A wrist strap protects parts from static; it does not make live electrical work safe.

Motherboards: the city map

VISUAL: Draw a green motherboard city with a CPU socket, parallel RAM slots, long expansion slots, and glowing copper roads.



Match the board, the case, and the parts!

FIT!

Board family	Typical maximum size	What changes
ATX	305 × 244 mm	More room for slots and connectors
microATX	244 × 244 mm	Smaller board; often fewer slots
Mini-ITX	170 × 170 mm	Compact; limited expansion space

• **Form factor.** The size, shape, and mounting layout. ATX means Advanced Technology Extended. ITX means Information Technology eXtended. Actual boards can be smaller than a family's maximum. [4]

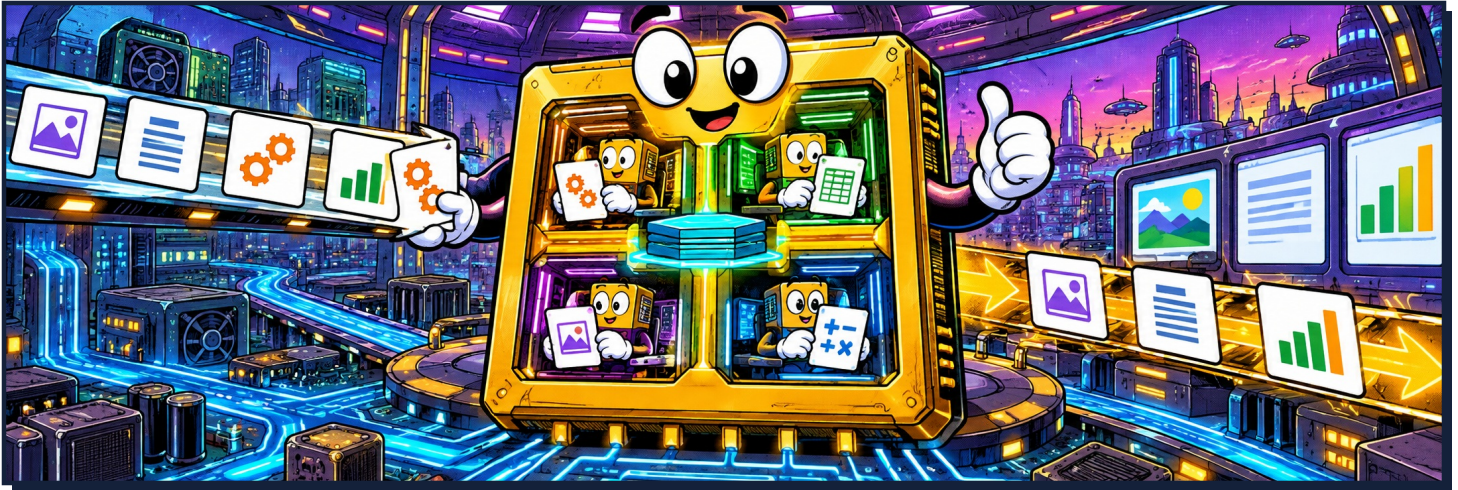
• **Chipset / bus.** A chipset helps manage platform connections. A bus is a path for data. Modern CPUs also contain controllers once found in separate motherboard chips.

• **Standoff.** A small spacer between the board and case. Install only the standoffs that match its mounting holes; an extra one can cause a short circuit. [5]

SIZE IS NOT COMPATIBILITY. Also check the CPU support list, firmware version, RAM type, power plugs, slots, case clearance, and cooler mount.

CPU: the instruction captain

VISUAL: Draw a gold processor captain with several worker stations and a tiny cyan cache shelf in a purple command center.



Fetch an instruction. Decode it. Do the work!

THINK!

- **Instruction.** One operation the processor knows how to perform. Fetch means get the instruction. Decode means work out its meaning. Execute means carry it out.

- **Core.** A processing engine inside a CPU. Multiple cores can work on different tasks when the software can share out the work.

- **Thread.** A stream of instructions. Some cores support multiple hardware threads. Two threads on one core are not the same as two complete physical cores.

- **Clock / GHz.** The clock sets a timing rhythm. GHz means gigahertz: billions of cycles per second. Higher GHz alone does not prove one CPU is faster than another.

- **Cache.** Small, fast memory near or inside the CPU. It keeps useful data close so the CPU does not always have to wait for slower memory. [6]

- **Architecture.** The CPU's instruction-set family. In common PC study use, x86 often means 32-bit and x64 means the 64-bit x86 extension. Arm is a different family; its historical name is Advanced RISC Machine.

- **RISC.** Reduced Instruction Set Computer. It describes an approach to instruction-set design. Apps and operating systems must support the processor architecture or suitable translation.

PERFORMANCE IS TEAMWORK. Core count, architecture, clock rate, cache, cooling, memory, and the workload all matter. Do not choose a processor from one number alone.

CPU fit: check more than the socket

VISUAL: Draw a robot matching a gold CPU to a motherboard using a blank compatibility checklist. Socket pins stay untouched.



Fits the hole? Now check the support list!

MATCH!

- **Socket.** The motherboard connector that holds the CPU. Match the exact supported processor family, not just the Intel or AMD brand. AMD means Advanced Micro Devices.
- **LGA - Land Grid Array.** The CPU has flat contact pads; spring contacts are in the socket. PGA, Pin Grid Array, places pins on the processor package. Both need careful alignment.
- **BGA - Ball Grid Array.** A chip package soldered to the board. Common in compact systems. It is not a routine socket upgrade.
- **Firmware support.** The board may need a compatible BIOS/UEFI version before a particular CPU will start. A shared socket does not guarantee support. Check the manufacturer's CPU list. [7]

- **Power and cooling.** Check board support, cooler capacity, mounting hardware, and space around the socket. TDP means Thermal Design Power; it is a cooling design figure, not always maximum electrical draw.
- **Multisocket.** A board with more than one CPU socket. It needs processors and a platform designed for that configuration. More cores in one CPU do not mean more sockets.
- **Install.** Use the model's alignment marks, retention method, and torque instructions. Never force a CPU into place. Support and protect the board while fitting the cooler.

COMPATIBILITY CHECK: socket + chipset/platform + CPU support list + firmware + power + cooler. A pin-count match is only one clue.

Cooling: move the heat away

VISUAL: Draw copper cooling fins, a blue fan, and a sealed liquid loop carrying orange warmth to a radiator.



Move heat from the chip into the room!

WHOOSH!

- **Heat sink.** A metal part that takes heat from a chip. Fins provide more surface area so air can carry the heat away.

- **Fan / airflow.** A fan moves air across hot parts. Intake brings cooler air in; exhaust pushes warmer air out. Check airflow arrows and the case plan instead of guessing.

- **Thermal paste.** A thin interface material that fills tiny gaps between the chip and cooler. It improves heat transfer; it is not glue. Use the cooler maker's amount and method. [8]

- **Thermal pad.** A shaped heat-transfer material. Thickness matters. Do not swap a pad for paste unless the manufacturer allows it.

- **Liquid cooling.** A pump moves coolant between a cold plate and a radiator. The radiator still needs airflow. An AIO, or All-in-One cooler, arrives as a preassembled loop.

- **Throttling.** The processor slows down to manage heat or power limits. This can make a machine seem weak even when its CPU is powerful.

- **Maintain.** Keep vents clear, remove dust safely, and check fan/pump connections. Monitor temperatures using the model's limits; there is no one safe temperature for every CPU.

EXAM CLUE: Shutdowns under load, loud fans, and high temperatures suggest checking cooling. Never run a CPU without its required cooler just to see what happens.

RAM: the temporary worktable

VISUAL: Draw green memory-stick workers sorting active task cards on a bright desk. A purple storage cabinet holds saved files.



More workspace can help a crowded task!

OPEN!

- **RAM - Random-Access Memory.** Working memory for active data and programs. Random access means the system can reach a location directly; it does not mean the data is random.
- **Volatile.** Needs power to retain contents. Ordinary system RAM is volatile; SSD and HDD storage are nonvolatile. Save work to storage before shutdown.
- **Capacity.** How much data fits. GB means gigabytes. More RAM helps when the current amount is too small, but unused extra RAM does not automatically speed up every task.
- **DDR - Double Data Rate.** A memory design that transfers data on both edges of a clock cycle. DDR3, DDR4, and DDR5 are different generations, not interchangeable upgrades. [9]

- **MT/s - Megatransfers per second.** Millions of transfers each second. DDR4-3200 names 3200 MT/s; its underlying clock is 1600 MHz. MHz means millions of cycles per second.
- **Channel.** A memory data path between the controller and RAM. Dual-channel operation uses two channels and can increase available memory bandwidth. Follow the slot-population guide.
- **Paging.** The OS can move some memory contents to storage when managing memory. Storage is much slower than RAM for this job; heavy paging can slow a busy system.

MEMORY HOOK: RAM = open worktable. SSD/HDD = saved-file cabinet. A larger cabinet does not automatically make the worktable larger.

RAM fit, errors & installation

VISUAL: Draw a long green desktop memory module and a short laptop module beside matching slots. Gold contacts stay clean.



Match the generation before clicking me in!

CLICK!

Generation	Desktop DIMM pins	SO-DIMM pins
DDR3	240	204
DDR4	288	260
DDR5	288	262

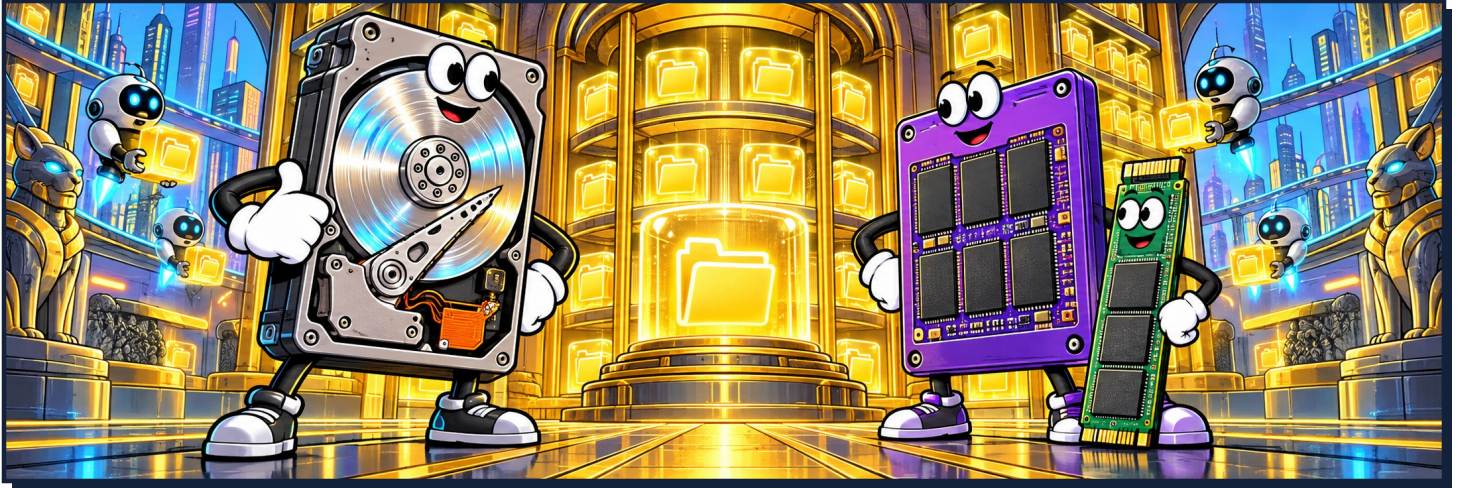
- **DIMM.** Dual In-line Memory Module: the familiar desktop/server memory module. SO-DIMM means Small Outline DIMM; it is shorter and common in laptops and small computers.
- **Pin count trap.** DDR4 and DDR5 desktop DIMMs both have 288 pins, but their electrical design and key notch differ. Never force one into the other generation's slot. [9]

- **ECC - Error-Correcting Code.** Supported ECC memory can detect and correct certain memory errors. The CPU, board, and module must support the right ECC type. Registered and unbuffered memory are not freely interchangeable.
- **DDR5 on-die ECC.** Corrects certain errors inside the memory chip. It does not, by itself, provide the same system-level protection as a platform using ECC DIMMs. [10]

INSTALL: Remove power, use ESD protection, align the notch, and seat the module as instructed. Match supported capacity, generation, module type, and slots; then verify detected memory.

HDD & SSD: two file keepers

VISUAL: Draw a silver HDD with spinning platters beside a purple SSD vault full of flash chips in a neon archive.



We keep saved files after the power stops!

SAVE!

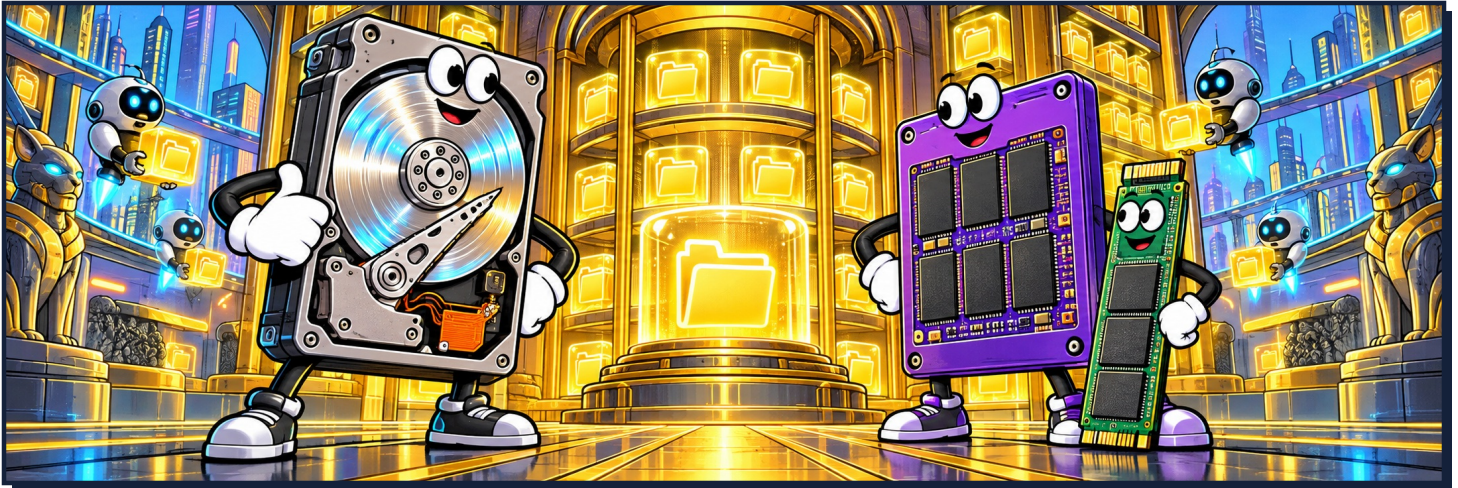
- **HDD - Hard Disk Drive.** Stores data magnetically on spinning platters. A moving head reads and writes. Mechanical movement adds delay and makes shocks risky during operation.
- **RPM - Revolutions Per Minute.** How many turns the platters make each minute. Common desktop examples include 5400 and 7200 RPM. RPM alone does not describe overall drive speed.
- **Drive sizes.** 3.5-inch HDDs are common in desktops; 2.5-inch drives fit many portable or compact systems. Those names describe drive form-factor families, not storage capacity.

- **SSD - Solid-State Drive.** Stores data in flash memory with no spinning platter or moving head. It usually responds to small random requests much faster than an HDD.
- **Capacity versus performance.** TB means terabytes. A 2 TB drive holds more data than a 1 TB drive, but capacity does not directly tell you latency or transfer speed.
- **Latency / IOPS.** Latency is how long a request waits. IOPS means Input/Output Operations Per Second: how many read/write operations a device completes under stated conditions.
- **Use cases.** An SSD works well for responsive startup and apps. HDDs can offer economical bulk storage. Compare endurance, interface, capacity, and workload, not just the box label. [11]

NO MOVING PARTS DOES NOT MEAN NO FAILURE. Back up important files from both HDDs and SSDs. A fast new drive is still only one copy.

Storage: shape, path & protocol

VISUAL: Draw a slim M.2 board and a boxed SATA drive choosing matching sockets on a purple motherboard road map.



My shape does not tell you my whole story!

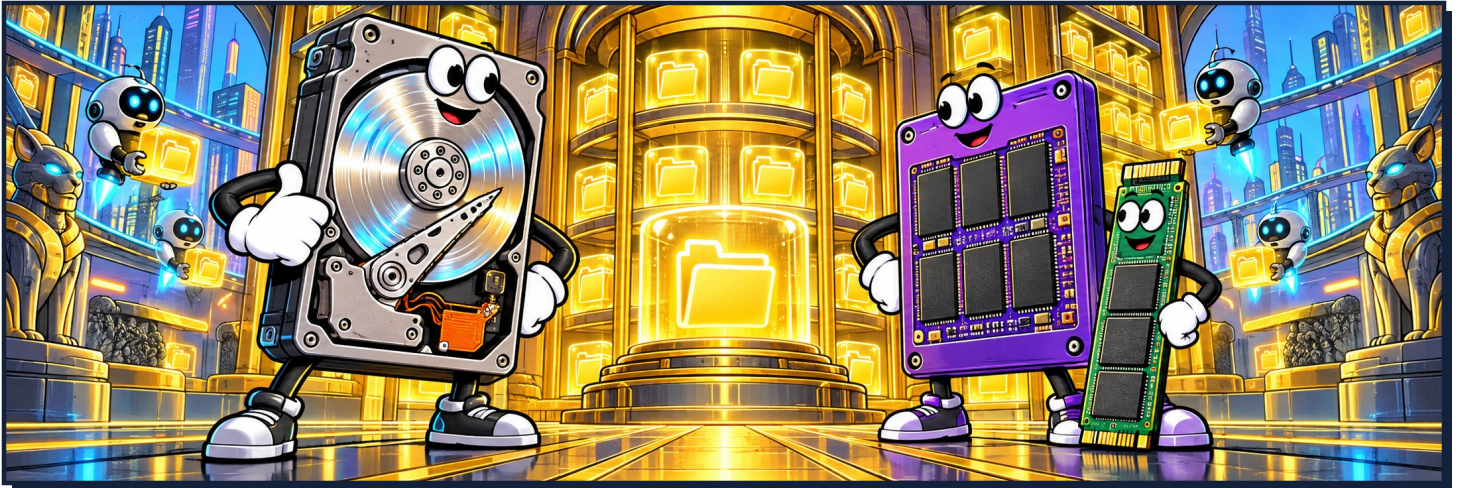
CHECK!

- **SATA.** Serial Advanced Technology Attachment. A storage interface used by many HDDs and SSDs. SATA 6 Gb/s is a link rate; real file transfers are lower.
- **NVMe.** Non-Volatile Memory Express. A storage command protocol designed for fast nonvolatile memory. Local NVMe SSDs commonly use PCIe connections.
- **PCIe.** Peripheral Component Interconnect Express. A fast serial connection arranged in lanes. Generation and lane count both affect available bandwidth.
- **M.2.** A compact module/socket form factor. M.2 SSDs may use SATA or PCIe/NVMe. The slot must support the drive's interface, keying, length, and boot requirements. [12]
- **2280.** An M.2 size: 22 mm wide and 80 mm long. It is not a speed rating. mSATA, mini-SATA, is an older small SATA form factor; it is not the same connector as M.2.
- **SAS.** Serial Attached SCSI. SCSI means Small Computer System Interface. SAS is common in enterprise storage. A normal SATA controller cannot operate a SAS drive.
- **Bandwidth units.** Gb/s means gigabits per second; GB/s means gigabytes per second. Eight bits equal one byte. Allow for protocol overhead before comparing a link rate with file-transfer speed.

EXAM TRAP: M.2 = form factor; NVMe = protocol; PCIe = connection path. A drive can fit physically and still be unsupported electrically.

Removable & optical storage

VISUAL: Draw pocket flash couriers, a memory card, and a shiny optical disc beside the purple archive vault.



Portable files still need protection!

PACK!

- **USB flash drive.** A small removable device using flash memory. USB means Universal Serial Bus. Its connector shape does not guarantee a particular data rate.

- **Memory card.** A removable flash format used in cameras and other devices. SD means Secure Digital; microSD is a smaller physical card format.

- **Reader compatibility.** Check both the card family and supported capacity standard. A physical adapter changes the fit; it cannot give an old reader unsupported capabilities.

- **Optical drive.** Uses a laser to read or write a disc. CD means Compact Disc. DVD means Digital Versatile Disc. Blu-ray is a name for another optical-disc technology.

- **R / RW.** R commonly means recordable: write once. RW means rewritable: erase and write again using a compatible format and drive. Blu-ray uses BD-R and BD-RE naming.

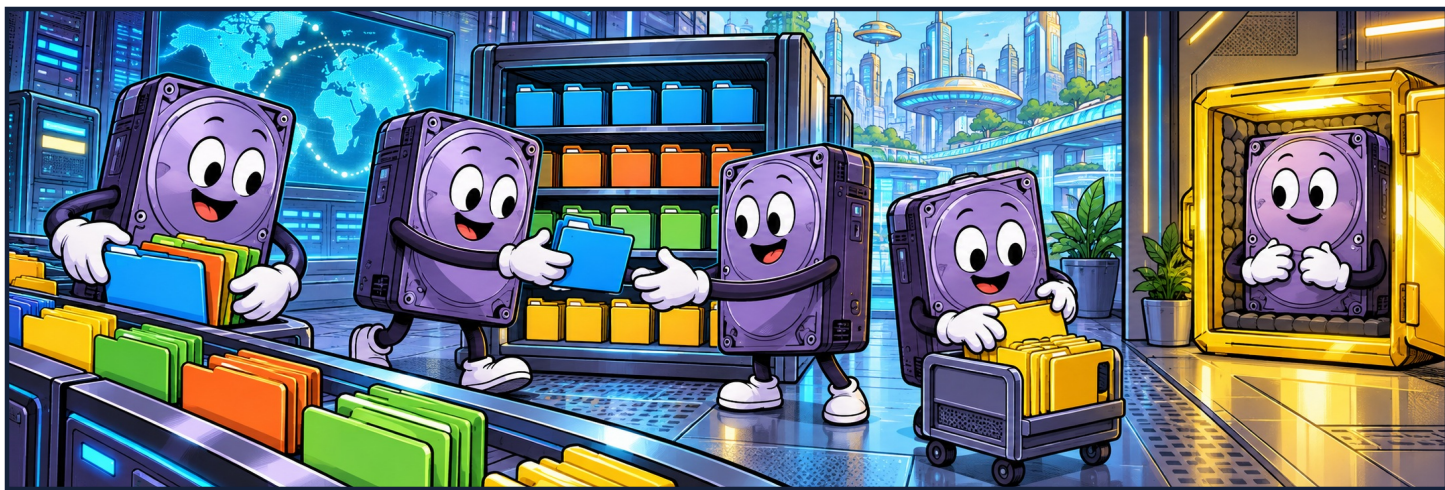
- **Typical capacities.** A common CD holds about 700 MB; a single-layer DVD, 4.7 GB; a single-layer Blu-ray Disc, 25 GB. Layers and formats change capacity. MB means megabytes. [13]

- **Use cases.** Move files, install approved software, or access older media. Follow the OS eject process before disconnecting removable storage when required; active writes must finish.

A BACKUP MUST BE A SEPARATE RECOVERABLE COPY. A flash drive can fail or disappear. Test restores and protect sensitive data on portable media.

RAID: the drive teamwork chart

VISUAL: Draw purple drive vaults sharing striped and mirrored folders. A separate gold box protects an independent backup.



Teamwork helps availability. Keep a backup too!

TEAM!

RAID	Typical minimum*	Usable capacity**	Drive failures tolerated
0: striping	2	$N \times S$	None
1: mirroring	2	S (two-drive mirror)	1 in a two-drive mirror
5: single parity	3	$(N - 1) \times S$	1
6: double parity	4	$(N - 2) \times S$	2
10: striped mirrors	4	$(N / 2) \times S$	1 per mirror pair

• **RAID.** Redundant Array of Independent Disks. Striping spreads data across drives. Mirroring keeps copies. Parity stores calculated recovery information. [14]

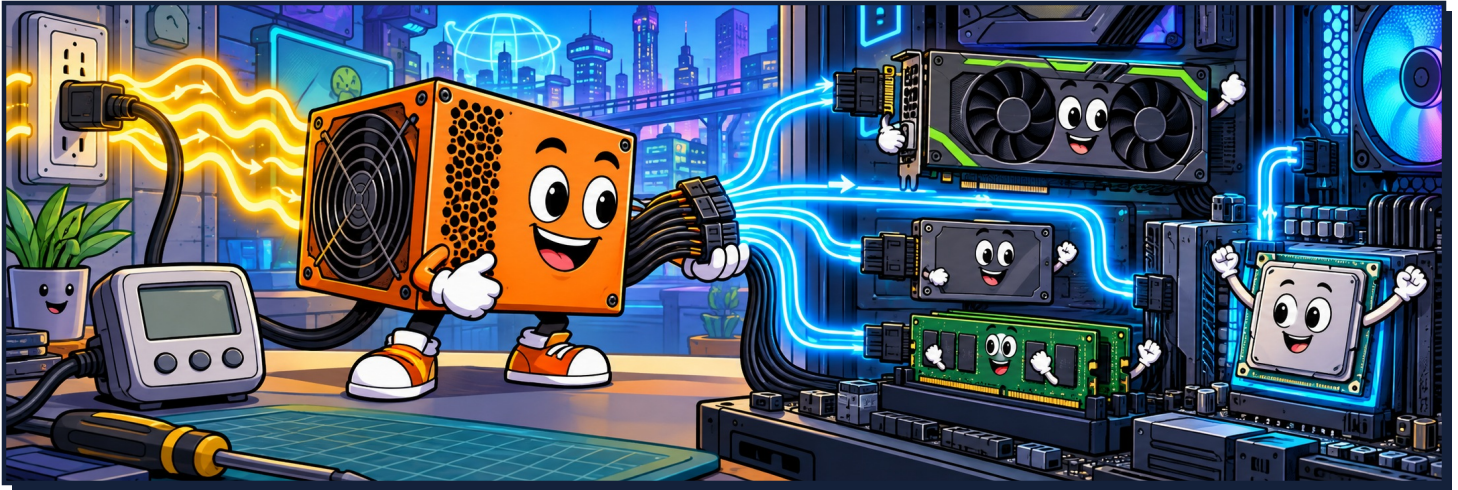
**N = number of drives; S = smallest drive capacity. These examples assume equal-size drives and ignore formatting/metadata. RAID 10 uses pairs; two failures in the same pair can destroy the array.

*Learn two drives for useful RAID 0 striping. Some controllers allow a one-disk RAID 0 virtual disk; that provides no striping benefit across drives.

RAID IS NOT BACKUP. It does not undo deletion, ransomware, or a building disaster. Replace the correct failed drive and follow the controller's rebuild procedure.

PSU: the power translator

VISUAL: Draw an enclosed orange power station converting wavy wall power into steady blue paths for the computer team.



Enough output, the right plugs, and room to grow!

ZAP!

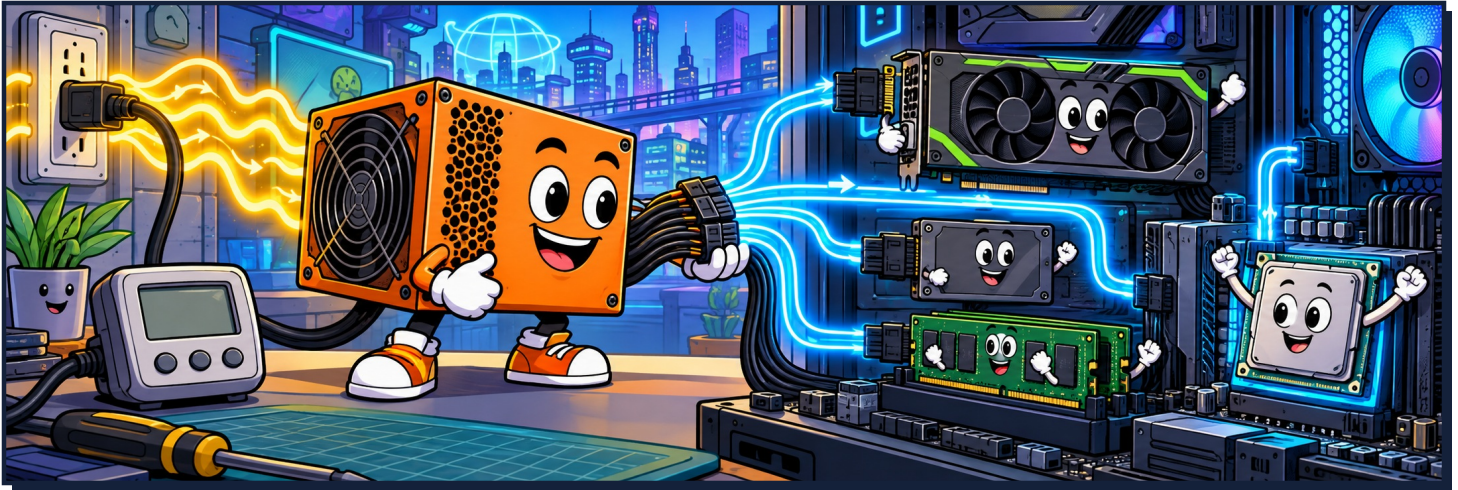
- **PSU - Power Supply Unit.** Converts wall AC, alternating current, into regulated DC, direct current, used by computer parts. Keep the PSU enclosure closed. [3]
- **Input voltage.** Common mains ranges are 110-120 VAC and 220-240 VAC. VAC means volts AC. Many modern units automatically accept a wide range; verify the label. Set any manual selector correctly before connecting power.
- **Output rails.** Common PC outputs include +3.3 V, +5 V, and +12 V DC. A rail is a power output at a specified voltage. V means volts; W means watts.

- **Wattage rating.** The PSU's supported output power under stated conditions. Add component needs, allow for load spikes and future upgrades, and check available connectors and rail limits.
- **Efficiency.** Useful output divided by power taken from the wall. If the PC needs 400 W and efficiency is 80%, wall input is about 500 W. About 100 W becomes heat. [15]
- **80 PLUS.** An efficiency certification with defined test conditions. The rating does not mean a 600 W PSU can supply only 480 W, and it does not promise the whole computer draws 600 W.

EXAM CLUE: Choose the right form factor, input range, output capacity, connectors, and quality. A bigger watt number cannot fix an incompatible cable.

Power plugs: never mix by looks

VISUAL: Draw orange power cables waiting at matching CPU, board, drive, and graphics-card gates. The station's case stays shut.



Similar plugs can carry different wiring!

STOP!

- **20+4-pin main power.** A motherboard power plug that can form a 24-pin connector. Use the exact connection required by the board; some vendor systems use proprietary power arrangements.

- **CPU power.** A separate 4-pin or 8-pin connector near the CPU socket; an 8-pin cable is often split 4+4. EPS12V is the common CPU power-connector standard name.

- **Graphics power.** Some cards need PCIe 6-pin or 6+2-pin power. Newer cards may use a 16-pin connector such as 12V-2x6. Follow the exact card and PSU instructions, including full insertion and bend clearance.

- **SATA / Molex power.** SATA power commonly has 15 contacts. A legacy 4-pin peripheral plug is often called Molex. Molex is a manufacturer name used informally for that connector family.

- **Modular.** Some or all cables detach at the PSU. This helps cable management. The PSU-side wiring is not universal, even when plugs fit. Use only cables explicitly compatible with that model. [16]

- **Redundant.** Multiple power supplies help keep a supported system running after one unit fails. This requires the right chassis and power design; it is not simply adding an extra loose PSU.

NEVER SWAP CPU 4+4 AND GPU 6+2 CABLES. Do not reuse modular cables just because the connectors look alike. Wrong wiring can destroy parts.

Expansion cards: add a new skill

VISUAL: Draw a cyan graphics-card artist, a sound-card musician, and a network messenger beside purple expansion slots.



Pick the card, the lane path, and the driver!

LEVEL UP!

- **Expansion card.** A circuit board that adds a function. The motherboard slot, physical room, power, firmware, and OS driver all must support it.
- **PCI / PCIe.** PCI means Peripheral Component Interconnect, an older expansion bus. PCI Express is newer and uses serial lanes. Their slots are not interchangeable.
- **x1, x4, x8, x16.** PCIe lane counts. A long physical slot may have fewer electrically connected lanes. Check the motherboard manual and shared-lane rules.
- **GPU - Graphics Processing Unit.** A processor well suited to graphics and many parallel calculations. A discrete GPU is separate hardware; an integrated GPU is part of another chip package.

- **Video card / VRAM.** A graphics card includes a GPU and supporting parts. VRAM means Video RAM: memory for graphics data. Dedicated VRAM differs from system RAM shared with integrated graphics.
- **Other cards.** A sound card handles audio. A NIC, Network Interface Card, connects a network. A capture card accepts a video source for recording or streaming.
- **Use case.** Choose a card to solve a stated need, such as more network ports or video capture. Install it safely, connect required power, load the correct driver, and test its function.

OUTPUT CLUE: With a discrete graphics card, connect the monitor to the intended graphics output. Motherboard video ports may not work if the CPU lacks integrated graphics.

Firmware: the startup gatekeeper

VISUAL: Draw a purple firmware chip checking the computer team before opening a gate for the operating-system robot.



Check the hardware. Find the boot program!

WAKE!

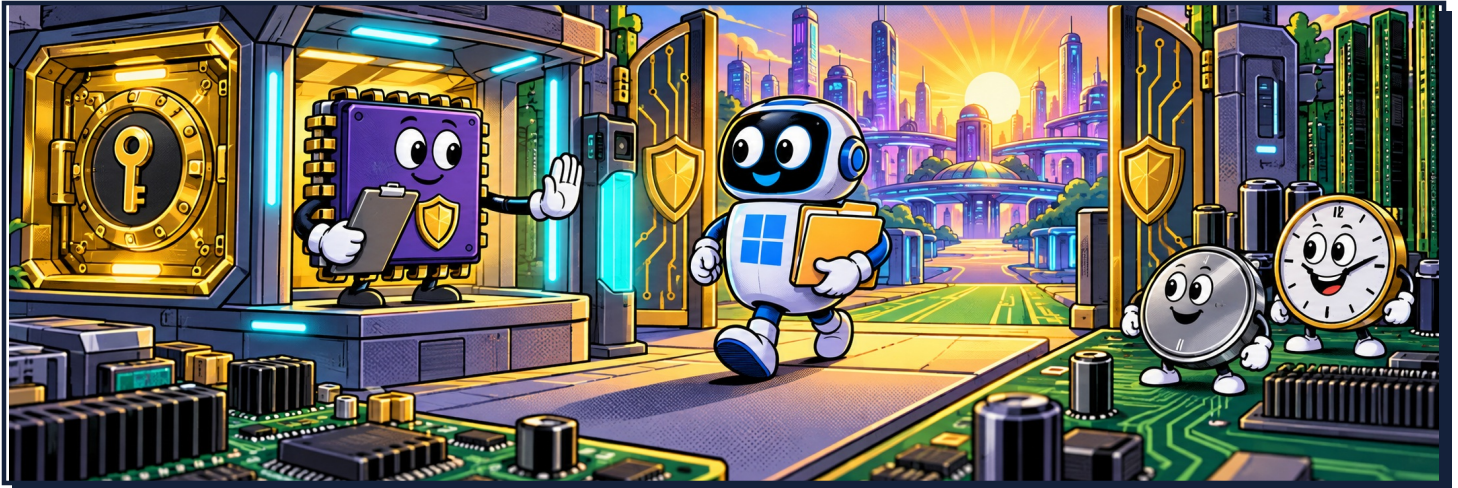
- **Firmware.** Low-level software stored in a device. It helps initialize and control hardware before and during normal use.
- **BIOS.** Basic Input/Output System, the traditional PC firmware system. People also say “BIOS setup” when they mean a modern firmware settings screen.
- **UEFI.** Unified Extensible Firmware Interface. A modern firmware interface with features such as richer boot management and Secure Boot support. [17]
- **POST.** Power-On Self-Test. Startup checks that help detect certain hardware problems. Beeps, lights, or codes must be read using the exact model’s documentation.

- **Boot order.** The preferred sequence of startup entries. Choose the intended OS boot manager or approved installation media. Detecting a drive is not the same as finding a working bootloader.
- **Settings.** Firmware may control USB permissions, fan curves, temperatures, and processor virtualization features. Record current settings before changing them.
- **Update safely.** Use firmware for the exact model/revision and the vendor’s update method. Keep power stable. Check encryption recovery requirements first; do not interrupt a flash update.

A DEAD CLOCK BATTERY IS NOT A DELETED OS. Lost time/settings can suggest a weak coin cell. Modern firmware itself is stored in nonvolatile flash, not kept alive by that battery.

Startup security & virtual machines

VISUAL: Draw the purple startup gate with a gold identity shield and a protected key vault beside the operating-system traveler.



Check trusted startup. Protect the keys!

GUARD!

- **Secure Boot.** UEFI checks digital signatures of boot software against trusted or blocked entries. It helps stop unauthorized startup code. It does not encrypt all files. [17]

- **TPM.** Trusted Platform Module. Provides protected cryptographic operations and key handling. It can support device identity, measured startup, and disk-encryption protection. It may be discrete or integrated. [18]

- **Encryption.** Transforms readable data into protected form using keys. A TPM can help protect keys, but it is not the same thing as the full-disk encryption software.

- **HSM.** Hardware Security Module. A specialized device for protecting and using cryptographic keys, often in business systems. It is not another name for every TPM.

- **Firmware versus boot password.** A setup password restricts firmware changes. A boot/power-on password restricts starting the system. Neither replaces OS account security or drive encryption.

- **Virtualization support.** Lets a hypervisor, the software managing virtual machines, use hardware assistance. Intel VT-x and AMD-V are familiar feature names. Enable only the supported setting needed for the task.

- **VM - Virtual Machine.** A software-defined computer running on a host. CPU support helps, but enough RAM and storage are also needed.

RECOVERY FIRST: Know your recovery keys and approved procedure before changing TPM or firmware settings. Clearing a TPM can affect access to protected data.

Displays: where the light comes from

VISUAL: Draw a backlit blue LCD beside an OLED whose colored pixels glow on their own in a purple art studio.



Backlight or glowing pixels? Know the difference!

GLOW!

- **LCD.** Liquid Crystal Display. Liquid crystals control how much backlight passes through. An “LED monitor” often means an LCD with an LED backlight.

- **LED.** Light-Emitting Diode. A small light source. Mini-LED uses many small LEDs for the backlight; it is not the same as self-emitting OLED pixels.

- **IPS.** In-Plane Switching, an LCD panel family often chosen for wide viewing angles and color consistency.

- **TN.** Twisted Nematic, another LCD family, often associated with fast response and narrower viewing angles. Actual performance depends on the model.

- **VA.** Vertical Alignment, an LCD family often associated with strong contrast. Contrast describes the difference between bright and dark areas. [19]

- **OLED.** Organic Light-Emitting Diode. Its pixels create their own light and can turn off for deep black. Repeated static images can cause uneven wear or burn-in.

- **Digitizer / inverter.** A digitizer senses touch or pen input. An inverter powered the fluorescent backlights in some older LCDs; modern LED backlights use different driver electronics.

EXAM TRAP: MINI-LED IS A BACKLIGHT TECHNOLOGY. IPS, TN, and VA describe LCD panel operation. OLED makes light at the pixel level. [20]

Pixels, refresh & color

VISUAL: Draw a gold magnifying glass over square pixels while a cyan monitor shows a smooth moving scene and a color rainbow.



Sharper, smoother, and more colorful are different!

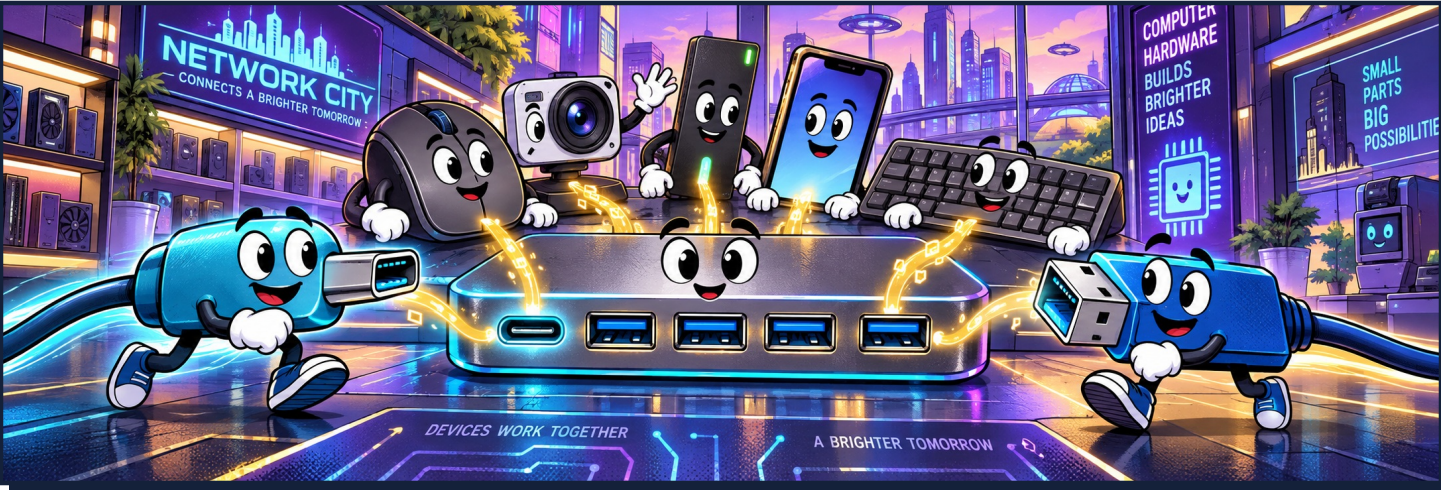
LOOK!

- **Pixel.** One small picture element. Resolution counts the pixel grid: 1920 × 1080 means 1920 across and 1080 down.
- **Native resolution.** The physical pixel grid of a fixed-pixel display. Other resolutions may need scaling, which can soften text or change image size.
- **Pixel density / PPI.** Pixels Per Inch describes how tightly pixels fit. Two screens with the same resolution can have different density if their sizes differ.
- **Refresh rate / Hz.** Hertz means cycles per second. A 144 Hz display can refresh up to 144 times each second in that mode. It does not force the computer to render 144 unique frames.
- **FPS / response time.** Frames Per Second measures rendered images. Pixel response time measures how quickly a pixel changes, often in milliseconds. Refresh rate, FPS, and response time are different measurements.
- **Color gamut.** The range of colors a display can reproduce. sRGB is a standard red-green-blue color space. Wide gamut alone does not guarantee accurate color; calibration also matters.
- **Choose the whole path.** The GPU, connector, cable, adapter, display, and settings must all support the requested resolution and refresh rate. Check model specifications. [21]

SCENARIO: A high-refresh screen stuck at 60 Hz may have a settings, cable, adapter, or output limit. Do not assume the display is broken.

USB: the shape is not the speed

VISUAL: Draw cyan USB-C and blue USB-A helpers guiding a keyboard, phone, camera, and flash drive into a neon hub.



Read the connector AND the data-rate label!

PLUG!

Specification naming	Maximum signaling rate	Memory hook
USB 2.0 High-Speed	480 Mb/s	The city bus
USB 3.0 / 3.2 Gen 1	5 Gb/s	The express train
USB 3.2 Gen 2	10 Gb/s	Two times the 5 Gb/s rate
USB 3.2 Gen 2×2	20 Gb/s	Two high-speed lanes

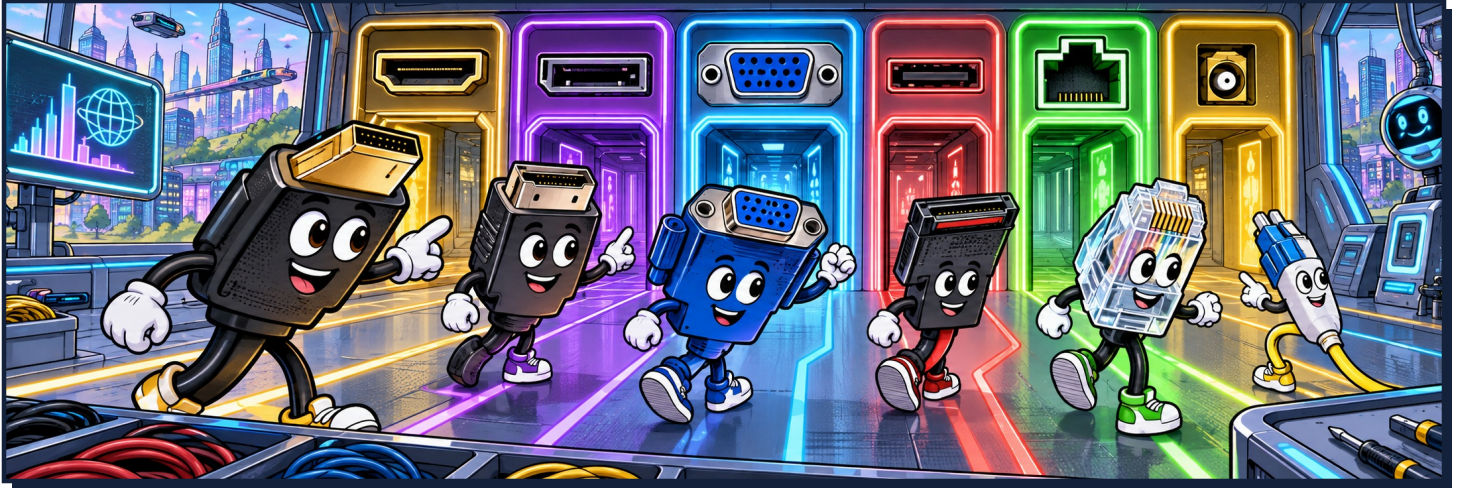
- USB - Universal Serial Bus.** Connects peripherals and can provide power. These are maximum link rates, not guaranteed file-copy speeds. All parts of the path must support the mode. [22]
- Connector shapes.** Type-A is the familiar rectangle; Type-B is common on older printers; Mini/Micro are smaller legacy families. USB-C is reversible and oval. Its shape alone proves neither speed nor video support.

- Power / distance.** USB Power Delivery can negotiate supported power. Check device, charger, and cable ratings. Classic USB 2.0 High-Speed passive segments are commonly limited to 5 m; faster modes have cable-specific limits.

EXTRA CONTEXT: USB4 and Thunderbolt can add higher rates and other capabilities. A USB-C cable can still be limited to USB 2.0 data. Never guess from color alone.

Video cables & Thunderbolt

●●● **VISUAL:** Draw HDMI, DisplayPort, VGA, and USB-C characters checking matching screen sockets in a purple cable workshop.



A matching shape still needs the right signal!

CONNECT!

- **HDMI.** High-Definition Multimedia Interface. Carries digital video and audio. Supported resolutions, refresh rates, and features depend on the whole connection, not simply the HDMI name. [23]

- **DisplayPort.** A digital display interface that can carry video and audio. Some USB-C ports support DisplayPort Alternate Mode; others do not. Alternate Mode uses the connector for another supported signal.

- **DVI.** Digital Visual Interface. DVI-D is digital, DVI-A is analog, and DVI-I integrates both signal types. Single-link and dual-link variants have different capabilities.

- **VGA.** Video Graphics Array. An older analog video connection commonly using a 15-pin connector. Analog-to-digital conversion usually needs an active adapter.

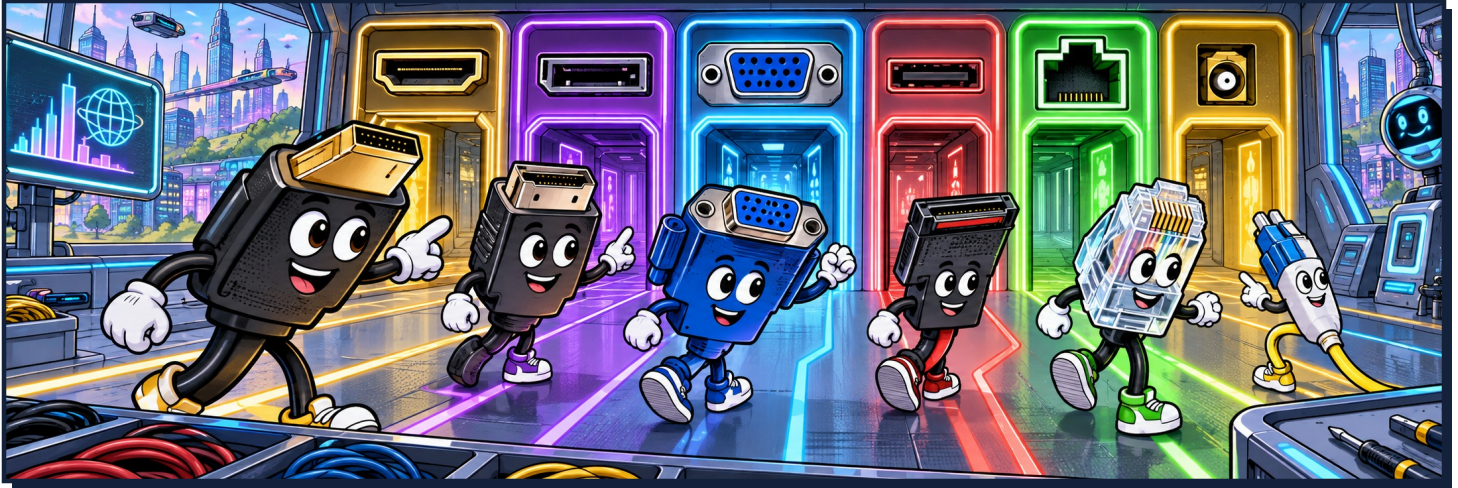
- **Thunderbolt.** A connection technology for data, displays, and peripherals. Versions 3 and 4 use USB-C and support up to 40 Gb/s. Original Thunderbolt 1/2 used the Mini DisplayPort connector shape. [24]

- **Version context.** Thunderbolt 5 supports 80 Gb/s bidirectional operation and can allocate up to 120 Gb/s in one direction for heavy display use. That is extra context, not a promise for every USB-C port.

ADAPTER CHECK: Confirm direction, signal conversion, power, resolution, refresh rate, and cable length. A passive shape adapter cannot invent an unsupported signal.

Internal cables & older connectors

VISUAL: Draw neat SATA and legacy connectors choosing matching gates. Gold screw trays and purple socket outlines keep the workbench organized.



Data, power, and control use different paths!

TRACE!

- **SATA data / power.** A typical SATA data connector has 7 contacts; SATA power has 15. A cabled SATA drive normally needs both. Internal SATA data cable length is commonly limited to 1 m. [25]

- **eSATA.** External SATA. Uses an external connector and cable system, commonly up to 2 m. Ordinary eSATA does not supply drive power; some powered variants do.

- **Headers.** Groups of motherboard pins or sockets for case switches, lights, USB, audio, or fans. Use the exact pinout: which signal belongs at each position.

- **Front panel.** The case power/reset switches and indicator lights connect to specified headers. LED polarity matters. A similar-looking header may carry a different voltage or function.

- **Serial / DB9.** A legacy serial port commonly uses a 9-pin D-sub connector, technically DE-9 but often called DB9. It can connect older instruments or console equipment. Match the electrical standard and settings.

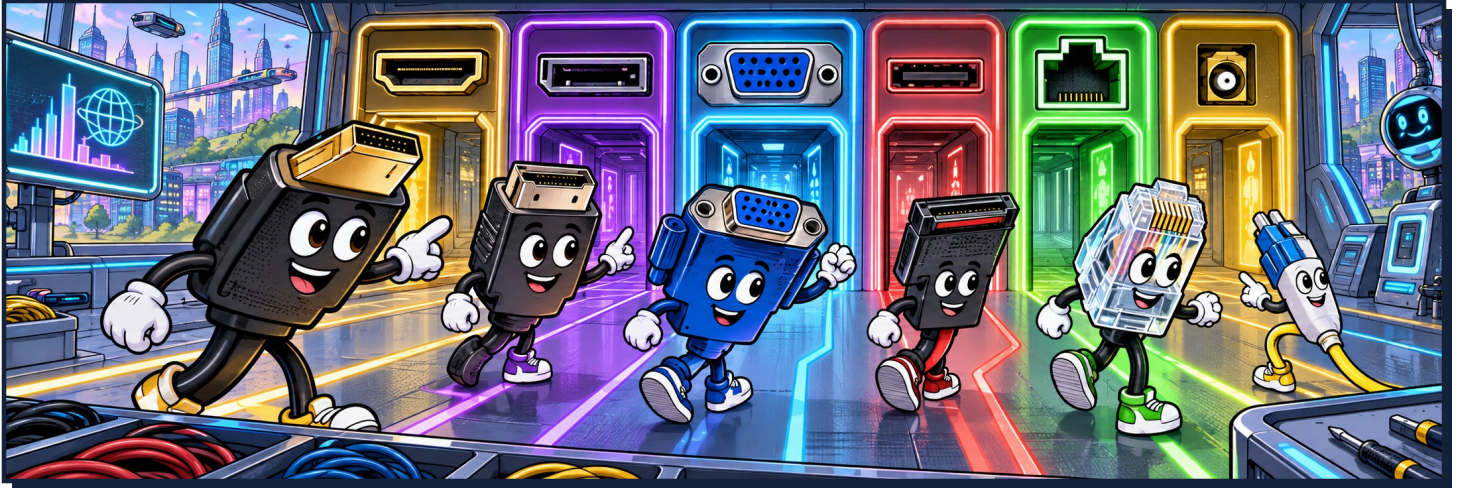
- **Lightning.** Apple's proprietary reversible connector used on many older Apple mobile devices. It is a connector name, not a synonym for USB-C or Thunderbolt.

- **Molex reminder.** The common legacy 4-pin peripheral power plug differs from SATA data and power. Use approved adapters only when the equipment and electrical ratings allow them.

DO NOT FORCE A CONNECTOR. Keying helps alignment but is not a full wiring guarantee. Power off before making internal connections unless the system explicitly supports that hot-swap operation.

Network cables: the hardware recap

VISUAL: Draw a clear RJ45 courier, a threaded coax helper, and a glowing fiber messenger beside bright neatly bundled cables.



Choose the cable for the signal and the place!

LINK!

- **Twisted pair.** Pairs of copper wires twist to help reduce interference. UTP means Unshielded Twisted Pair; STP means Shielded Twisted Pair. Cat means category, a performance classification.

- **Common reach.** A standard copper Ethernet channel is often 100 m total, typically 90 m permanent cable plus patch cords. Cat 5e commonly supports 1 Gb/s at 100 m; Cat 6A supports 10 Gb/s at 100 m. [26]

- **RJ45 / RJ11.** RJ means Registered Jack. Ethernet's 8-position modular plug is commonly called RJ45. RJ11-style connectors are smaller and commonly used for telephone service.

- **T568A / T568B.** Two recognized wire-pair arrangements for terminating twisted-pair cabling. Use the correct scheme consistently for the intended cable and verify it with a tester.

- **Coax / F-type.** Coaxial cable has a center conductor, insulation, and surrounding shield. A threaded F-type connector is common for cable internet and television.

- **Fiber / connector names.** Single-mode fiber carries one light mode; multimode carries several. Common connectors include ST, Straight Tip; SC, Subscriber Connector; and LC, Lucent Connector. Reach depends on the optics and fiber.

INSTALLATION RATINGS MATTER: Plenum cable meets specified fire/smoke requirements for air-handling spaces. Direct-burial cable handles underground conditions. Neither label alone tells you data speed. Never look into active fiber.

 VISUAL: Draw a robot assembling a planned desktop from compatible parts. A gold checklist tracks fit, power, cooling, and testing.



- UPGRADE CLUE:** A nearly full drive calls for storage work; heavy paging may call for more RAM. Diagnose the bottleneck before buying a faster CPU.

Printers: setup and first connection

VISUAL: Draw a purple multifunction printer being installed on a sturdy desk beside a cyan computer. Packing tabs sit in a nearby tray.



Place me well. Connect me. Choose my driver!

PRINT!

- **MFD / MFP.** Multifunction Device / Multifunction Printer. Combines tasks such as printing, scanning, and copying. Some models also fax.

- **Unbox and place.** Follow lifting instructions, remove shipping restraints, and allow ventilation and service clearance. Check electrical needs and keep the device within rated temperature/humidity limits.

- **Connectivity.** USB can connect one nearby host. Ethernet or Wi-Fi can put the device on a network. Use approved network settings and verify the assigned IP address.

- **Driver.** Software that helps the OS send supported jobs. Choose the model, operating system, and required features; a driver that prints basic text may not expose every tray or finishing option.

- **PCL / PostScript.** PCL is Printer Command Language, also called Printer Control Language in some study lists. PostScript is another page-description language. Match the language supported by the printer and application needs. [27]

- **Firmware / first test.** Use approved firmware updates for the exact model. Print a device configuration page, then an OS test page. That helps separate printer operation from connection/driver issues.

EXAM CLUE: A network-connected printer does not automatically have the right driver, address, permissions, or paper settings. Check each part of the print path.

Print settings, scanning & security

VISUAL: Draw a gold page waiting inside the purple printer until a worker taps a badge. The feeder and flatbed scan different documents.



Right paper. Right person. Right destination!

RELEASE!

- **Duplex / orientation.** Duplex prints on both sides; simplex prints on one. Portrait is tall page orientation; landscape is wide. Match the job's page size and layout.

- **Tray / quality.** Select the correct paper size and type. DPI means Dots Per Inch, a print-resolution measure. Higher quality can take longer and use more supplies.

- **Queue / spooler.** A print queue lists waiting jobs. A spooler manages stored print jobs for delivery. A print server can share printers and centralize management.

- **Secure print.** A job waits for a PIN, personal identification number, badge, or other authentication at the printer. Authentication checks the user; permissions control allowed actions.

- **Audit log.** A record of events such as who released a job. It can help an organization track use and investigate problems.

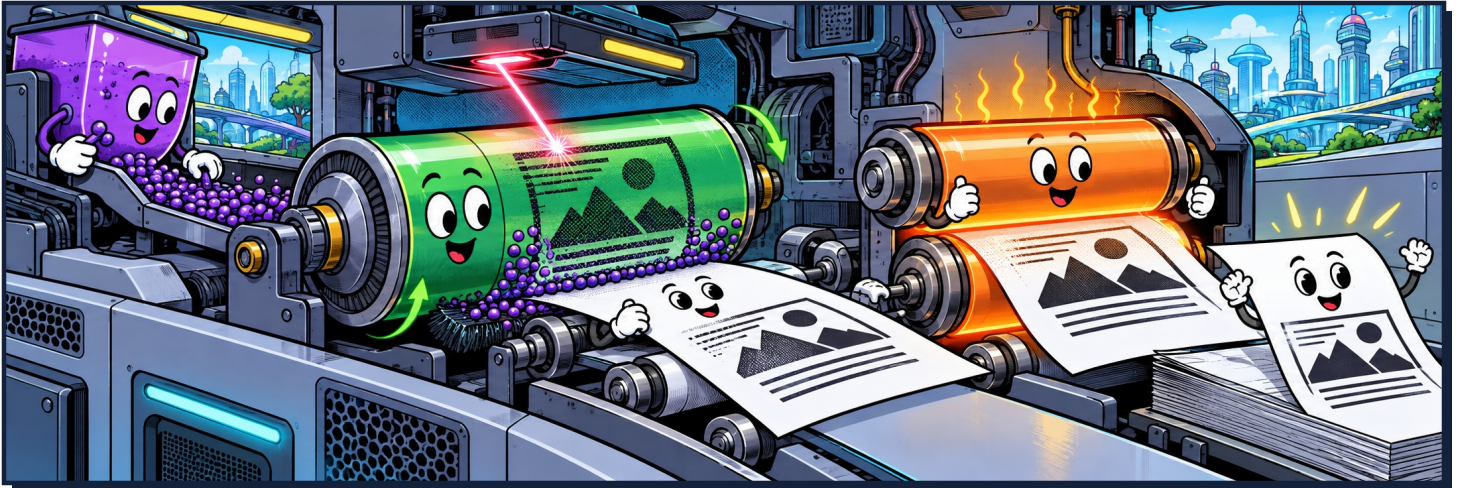
- **ADF / flatbed.** An Automatic Document Feeder moves a stack through a scanner. A flatbed holds the original on glass, useful for books or fragile material. Not every ADF scans both sides automatically.

- **Scan destinations.** Email, approved cloud services, or an SMB share can receive scans. SMB means Server Message Block. Verify the destination, account permissions, and network configuration. [28]

USE CASE: For private documents on a shared printer, use supported held-job release. Do not leave sensitive pages sitting in an open output tray.

Laser printing: the seven-step factory

VISUAL: Draw a neon laser-printer factory with a green drum, purple toner, traveling paper, and orange fuser rollers.



Draw with charge. Add toner. Bond it to paper!

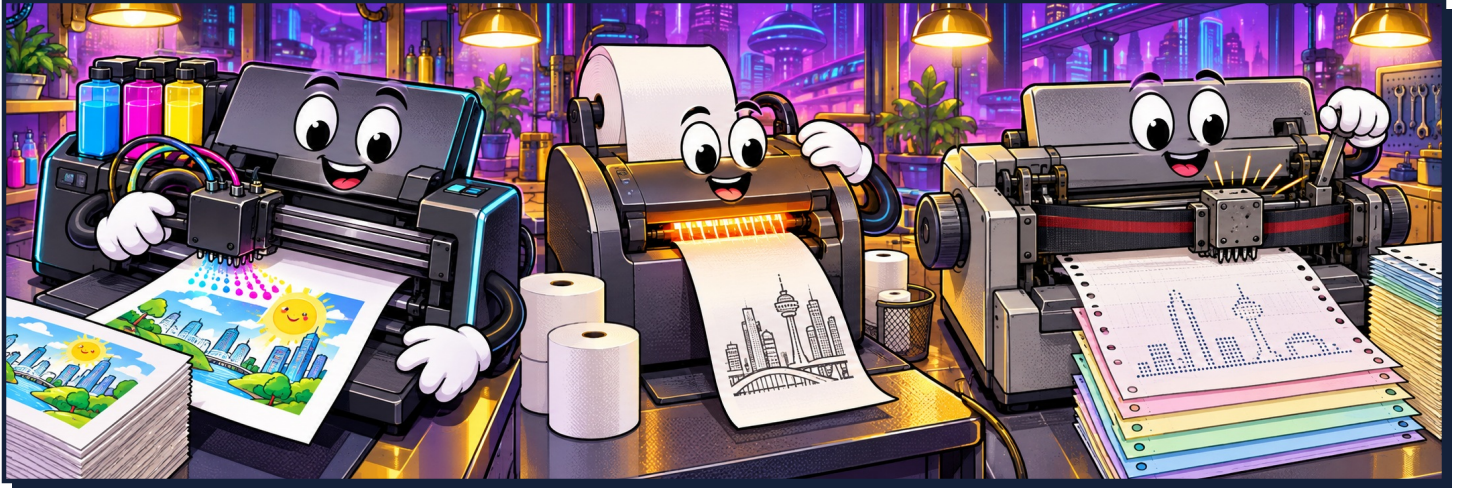
FUSE!

- **1. Processing.** The printer turns the job into a page image it can print. Think of preparing the factory's work plan.
- **2. Charging.** A charge roller or other charging part gives the photosensitive drum a uniform electrical charge.
- **3. Exposing.** The laser changes charge on selected drum areas, creating an invisible image pattern.
- **4. Developing.** Toner, a fine printing powder, is attracted to the intended image areas. Toner is not liquid ink.
- **5. Transferring.** The toner image moves to the paper, directly or through a transfer system, depending on printer design.
- **6. Fusing.** Heat and pressure bond toner to the page. The fuser is hot; it is not the same part as the imaging drum. [29]
- **7. Cleaning.** Leftover toner is removed from the drum so it can be prepared for another image. Several stages occur continuously as the parts rotate.

MEMORY CHAIN: Process → Charge → Expose → Develop → Transfer → Fuse → Clean. Toner that rubs off can suggest a fusing or paper-setting problem; diagnose before replacing parts.

Inkjet, thermal & impact teams

VISUAL: Draw three stations: tiny colorful ink droplets, a thermal receipt roll, and pins striking a ribbon onto layered paper.



Different print jobs need different methods!

TAP!

- **Inkjet.** A printhead places tiny liquid-ink droplets on paper. Cartridges or tanks supply ink. Use cases include color documents and photos on suitable media.

- **Nozzles / alignment.** Nozzles are tiny ink openings. Clogged nozzles can leave missing lines. Alignment helps droplets land where they belong.

- **Direct thermal.** A heated printhead darkens specially coated paper. Common for receipts. It needs the correct thermal side facing the head and does not use a normal ink cartridge. [30]

- **Thermal transfer.** A related technology melts material from a ribbon onto media. Do not assume every printer called thermal uses no ribbon.

- **Impact / dot matrix.** Pins strike an inked ribbon against paper to form dots. Physical impact can make copies through multipart paper. It is useful for forms that need duplicate layers.

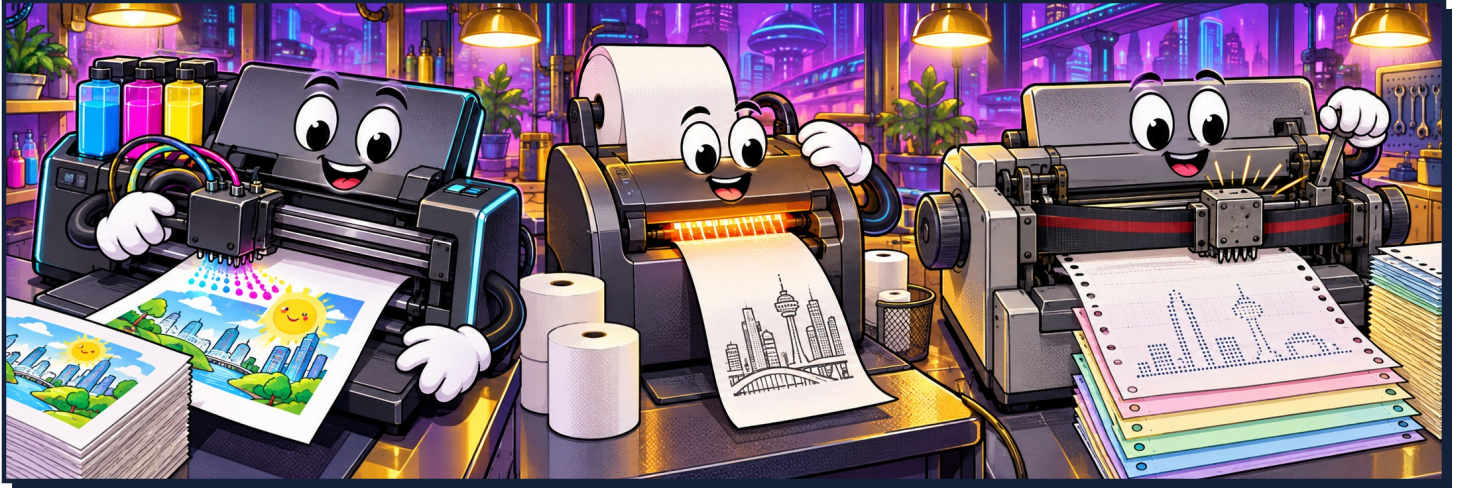
- **Multipart paper.** Stacked sheets designed to copy the impact onto lower layers. An ordinary inkjet or laser printer cannot make those copies by striking the stack.

- **Consumables.** Supplies used up during printing: ink, toner, ribbons, or media. Match the device and job; similar-looking cartridges or paper can be incompatible.

MATCH THE CLUE: Photos → often inkjet. Receipts on coated paper → direct thermal. Multipart forms → impact. High-volume crisp office pages → often laser.

Printer maintenance: clean with care

VISUAL: Draw the printer crews beside correct replacement supplies and a tidy cleaning kit. Hot heads cool before service.



Follow my service steps, then print a test!

CLEAN!

- **Laser.** Replace toner and scheduled maintenance-kit parts as directed. Kits may include rollers or a fuser. Use the printer's cleaning and calibration routines. [31]
- **Toner safety.** Avoid spreading fine powder. Use the manufacturer's spill procedure and only equipment rated for toner; do not use an ordinary household vacuum. Let hot parts cool before handling.
- **Inkjet.** Run a nozzle check first. Use the approved head-cleaning routine only as needed; repeated cleaning consumes ink. Replace the correct cartridge/tank supply and align when required. [32]
- **Thermal.** Remove power and let the head cool. Clean the heating element with the manufacturer-approved materials. Load the correct paper and remove debris from the feed path. [30]
- **Impact.** Replace the ribbon, paper, and printhead according to the manual. Set the correct paper path and head gap for the forms. Never work around moving parts.
- **Jams and rollers.** Use the designated access doors and pull paper as directed. Remove scraps without scratching delicate surfaces. Repeated jams may involve wrong media, worn rollers, or a feed problem.
- **Calibrate / verify.** Calibration adjusts output toward a known target. After maintenance, test the affected function and record what was changed.

NO UNIVERSAL CLEANER: The safe liquid, swab, cloth, or vacuum depends on the component. Keep fingerprints off imaging surfaces and follow the exact model's instructions.

Fault clues: diagnose before replacing

VISUAL: Draw a robot detective using a gold checklist beside an open computer. A cool blue fan and purple drive wait for tests.



A symptom is a clue, not a guilty verdict!

TRACE!

- **No power.** Start with the outlet, cable, switch, and connections. Check main and CPU power plugs. A fan that twitches is not proof that every PSU output is healthy.
- **No display.** Check the monitor's power/input, cable, intended graphics output, and POST indicators. A black screen does not always mean a failed GPU.
- **POST code / memory fault.** Use the model's code guide. With power removed, check supported RAM and seating. Change one thing at a time so the result means something.

- **Overheating / shutdown.** Check vents, dust, fan/pump operation, cooler contact, and temperature readings. Stop using a system that smells burned or shows electrical damage.
- **Drive warning.** S.M.A.R.T. means Self-Monitoring, Analysis and Reporting Technology. A warning deserves attention, but passing health data is not a guarantee. Protect important data before stressful tests.
- **Boot device missing.** Check detection, connections, firmware settings, and the boot entry. Do not format a drive merely because the OS does not start.
- **Print failure.** Check the device's own test page, media, queue, driver, and connection. Separate a job problem from a feed or marking problem.

WORKFLOW: Identify → form a theory → test → fix safely → verify → document. It is a useful support practice; the official objectives distinguish it from a separately tested methodology topic. [1]

Quick recall: words and units

●●● **VISUAL:** Draw the neon hardware team lining up with a gold checklist: processing, workspace, storage, power, and output.



Say what it means, then say what it does!

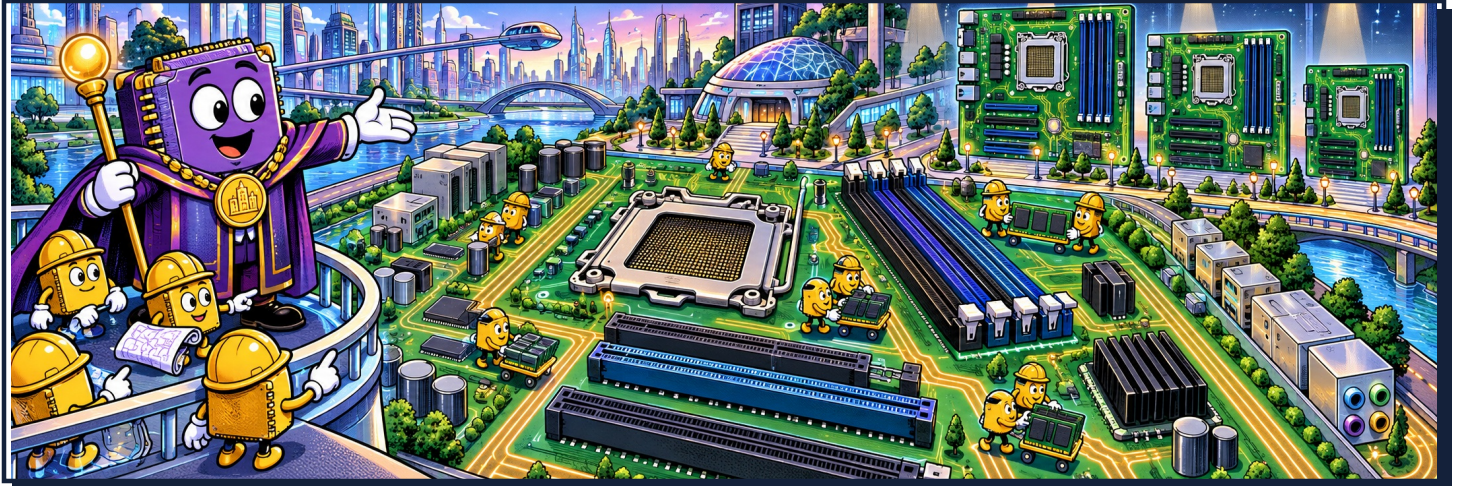
RECALL!

Clue	Term	Meaning / distinction
Runs instructions	CPU	Central Processing Unit
Temporary work area	RAM	Random-Access Memory
Saved magnetic data	HDD	Hard Disk Drive
Saved flash data	SSD	Solid-State Drive
Graphics calculations	GPU	Graphics Processing Unit
Wall power conversion	PSU	Power Supply Unit
Million transfers/s	MT/s	Memory data rate; not MHz
Billions of cycles/s	GHz	Clock frequency; not total performance
Bits versus bytes	b / B	8 bits = 1 byte
Screen refreshes/s	Hz	Refresh rate; not rendered FPS

FINAL MATCH: M.2 is a shape. NVMe is a protocol. PCIe is a connection path. RAID is an array strategy. Backup is a separate recoverable copy.

Sources & exam alignment

VISUAL: Draw a neat neon reference desk with checked gold cards beside the hardware team.



Check the facts behind the colorful story!

CHECK!

Exam map: 3.1 displays; 3.2 cables; 3.3 RAM; 3.4 storage; 3.5 boards/CPUs/cards; 3.6 power; 3.7 printer setup; 3.8 printer maintenance. [1]

1. CompTIA: Core 1 220-1201 objectives
2. Dell: Electrical and ESD safety
3. Corsair: PSU manual and safety
4. ASUS: Motherboard dimensions
5. ASUS: Motherboard installation
6. Intel: CPU specifications and features
7. Intel: CPU / chipset / firmware support
8. Intel: Thermal interface material

9. Crucial: DDR memory generations
10. Kingston: DDR5 memory standard
11. Kingston: SSD and HDD differences
12. Kingston: SATA / M.2 / NVMe FAQ
13. Sony: Optical disc capacity
14. Dell: RAID levels and specifications
15. Corsair: PSU efficiency ratings
16. Corsair: Modular cable compatibility

Reviewed September 18, 2026. Independent study material; not endorsed by CompTIA or Per Scholas. Technical labels are separate from the fictional artwork.

Sources: displays, cables & printing

VISUAL: Draw a neat neon reference desk with checked gold cards beside the hardware team.



Check the facts behind the colorful story!

CHECK!

Reading note: Source numbers in the lessons point to these clickable primary references. Model-specific manuals always govern installation and maintenance.

17. **Microsoft:** Secure Boot

18. **Microsoft:** TPM fundamentals

19. **BenQ:** LCD panel types

20. **Display technology** reference

21. **Dell:** Monitor specifications

22. **USB-IF:** USB 3.2 data rates

23. **HDMI:** Cable types and capabilities

24. **Intel:** Thunderbolt technology

25. **StarTech:** SATA cable specifications

26. **CommScope:** Category 6A cabling

27. **HP:** Choosing printer drivers

28. **HP:** Scan to network folder

29. **HP:** How laser printers work

30. **Epson:** Thermal printhead cleaning

31. **HP:** Laser print-quality maintenance

32. **Epson:** Inkjet maintenance

Original illustrations created with the built-in image-generation tool. Characters, glowing paths, and factory scenes are memory aids, not literal wiring or service diagrams.

STAR QUIZ TIME

VISUAL: Draw the CPU, RAM, storage, and printer heroes cheering under three bright gold stars in the neon computer city.



Your turn. Pick the part and explain why!

STAR!

- 1. The crowded worktable:** A computer slows while many apps run. Memory use is high and the OS keeps moving data to storage. Which resource might need an upgrade: RAM or drive capacity? What must you check before installing it?
- 2. The tiny drive trap:** A new M.2 NVMe SSD fits a slot that supports only M.2 SATA. Will it work just because it fits? Explain M.2, NVMe, and PCIe.
- 3. The rubbing toner:** A laser-printed image rubs off the paper. Which printing stage normally bonds the toner? Name a setting to check and one service-safety step.

ANSWERS - CHECK YOUR HERO SKILLS!

- 1. RAM may be the bottleneck.** Nice diagnosis! Check supported generation, capacity, module type, and slot arrangement. Confirm the cause first; adding storage capacity does not increase RAM.
- 2. No.** Great catch! M.2 describes the form factor; NVMe is a storage protocol commonly carried over PCIe, the connection path. A SATA-only M.2 slot does not support that NVMe drive.
- 3. Fusing.** You found the right stage! Check the paper type/weight setting and media, then follow diagnostics. Let the fuser cool and follow the manual before service. A symptom alone does not prove the fuser must be replaced.